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**Amrita School of Computing, Amritapuri**  
**BTech - Computer Science and Engineering (AI)**

22AIE311 -Software Engineering (Project based)

**LAB SHEET 1**

| CO   | Course Outcomes                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------|
| CO01 | Understand the basic principles of Software Engineering.                                                          |
| CO02 | Understand how to choose the appropriate SDLC models depending on the user requirements.                          |
| CO03 | Understand the concept of Requirements Engineering.                                                               |
| CO04 | Understand the concept of Software Design.                                                                        |
| CO05 | Understand how to apply the knowledge, techniques, and skills in the development of software and its maintenance. |

***Requirement Engineering – Inception and Elicitation Phase***

***Objective***

To understand and perform the Inception and Requirement Elicitation activities of Requirement Engineering for a given software project.

**PART A – INCEPTION PHASE**

**Task 1: Problem Identification**

[CO03]

1. Define the problem clearly.

Contemporary users subscribe to various digital solutions, including streaming platforms, cloud solutions, productivity, and online learning platforms. These services are consumed on several gadgets such as smart phones, laptops, and desktops. Nonetheless, end-users do not have a unified platform to evaluate the cost of their subscriptions,

the usage of apps on different devices, and to assess whether they make good use of such subscriptions.

Consequently, users are struggling to deal with how they are spending their digital time, whether their subscriptions are worth their money, and the impact of such services on their productivity and finances.

The suggested system tries to combine the subscription cost tracking, cross-gadget usage following, and clever time optimization into one platform.

## 2. What existing issues does the system solve?

The following issues are solved by the system:

1. Absence of centralized subscriptions:  
Users subscribe to numerous services and fail to monitor recurrent payments.
2. Lack of visibility of digital patterns of use.  
The screen time is not aggregated and is counted individually on each device.
3. Financial inefficiency  
Subscriptions are usually purchased by the users who do not consume them at all.
4. No relationship between expenditure and consumption.  
Existing systems track either expenses or screen time, but not both together.
5. Inefficient time management on online activities  
Users have difficulty in striking a balance between productivity applications and entertainment.
6. Absence of astute thinking.  
The majority of the tools lack tools for analytics like cost-per-hour usage or the subscription ROI.

## Task 2: Identify Stakeholders

| Sl. No | Stakeholder Name | Role                               | Interest in System                                                                        |
|--------|------------------|------------------------------------|-------------------------------------------------------------------------------------------|
| 1      | ADITHYA RAJESH   | Frontend & Integration Developer   | Develops the web interface, dashboards, and integrates device monitoring modules          |
| 2      | BALU M KRISHNA   | Backend & Data Engineer            | Responsible for backend development, database design, and analytics processing            |
| 3      | ADITHYAN R       | Project Manager / System Architect | Oversees system design, architecture planning, and ensures the project meets requirements |

| Stakeholder          | Role                                    | Interest                                           |
|----------------------|-----------------------------------------|----------------------------------------------------|
| End Users            | Individuals managing sub-<br>scriptions | Want better control of expenses<br>and screen time |
| System Administrator | Maintains servers and de-<br>ployment   | Ensures reliability and system per-<br>formance    |

### Task3: Define System Scope

#### 1. What the system will do

##### The system will:

- Provide subscribers with the opportunity to create, maintain and monitor digital subscriptions.
- Monitor Android and windows usage of applications.
- Filed and processed data on data stores and usage in a centralized database.
- Create analytics dashboards to be used and spent.
- Calculate metrics such as:
  - Cost per hour usage
  - Subscription utilization
  - Return on investment (ROI)
- Give physical reports and usage reports.
- Recommend the best time periods to be allocated to productivity and entertainment applications.

Give users the option of establishing objectives and the timeframes.

#### 2. What the system will NOT do

##### The system will not:

- Regulate the application usage itself.
- View or edit application-based user content.
- Substitute the current subscription services.
- Offer payment gateway or financial transactions.
- Cyberspy on the devices without authorization.

### 3. System boundaries

#### **The system includes:**

- Web-based dashboard
- Backend analytics server
- Database storage
- Android surveillance software.
- Windows desktop surveillance agent.

#### **External systems include:**

- Operating systems of devices (Android /Windows).
- Cloud hosting platforms
- Email/ notification services.
- The system deals with these components and does not govern them.

### **Task 4: Identify Constraints**

#### 1. Budget constraints

It is an academic project that has few financial resources. Thus, the system will be based on the open-source technologies and free-tier cloud services.

#### 2. Time constraints

The project should be done within the term of the academic semester, which consists of design, development, testing, and documentation.

#### 3. Technical constraints

- The monitoring of the devices should be in conformity to Android and Windows system APIs.
- Privacy of data should be ensured.
- The synchronization of cross-device data should be effective.
- The system should be able to support the real-time logs of usage without heavy load on the server.

#### 4. Legal constraints

**The system must comply with:**

- User privacy policies
- Data protection standards
- Monitoring devices with definite user agreement.
- Collection of sensitive personal data should be avoided.

### **Task 5: Feasibility Study**

#### 1. Technical Feasibility

The project is technically feasible because the required technologies are well established. Application usage tracking can be implemented using Android usage statistics APIs and Windows system monitoring libraries. Modern frameworks such as React, Node.js, and PostgreSQL provide reliable infrastructure for building scalable web applications.

#### 2. Economic Feasibility

The project is economically feasible since it relies primarily on open-source technologies. Free or low-cost cloud platforms such as AWS free tier, Vercel, or Google Cloud can be used for deployment. Therefore, the overall development cost remains minimal.

#### 3. Operational Feasibility

The system is designed to be user-friendly and accessible through a web interface. Users can easily manage subscriptions, view analytics, and monitor device usage without requiring technical expertise. Therefore, the system is practical for everyday use.

#### 4. Schedule Feasibility

The development tasks can be completed within the academic semester timeline by dividing work among team members. Each module—frontend, backend, device monitoring, and analytics—can be developed in parallel.

**Task 6: Select Elicitation Techniques**

Choose at least **two techniques**:

**1. Brainstorming**

Brainstorming is used to generate ideas and identify possible features of the system through collaborative discussion among team members. During brainstorming sessions, the development team discusses user problems related to subscription management, screen-time monitoring, and productivity tracking.

This technique helps in identifying innovative solutions such as subscription ROI calculation, cost-per-hour usage analysis, and intelligent time allocation. Brainstorming also allows the team to explore multiple design ideas and determine which features are most valuable for the users.

**2. Observation**

Observation involves studying how users interact with digital devices and subscription services in their daily routines. By observing user behavior, developers can understand how much time users spend on different applications, how frequently they use their subscriptions, and whether they track their expenses effectively.

This technique helps in identifying real-world problems such as unused subscriptions, excessive screen time, and inefficient time allocation between productivity and entertainment applications. Observation therefore provides realistic insights that help design a system that accurately monitors usage patterns and provides meaningful recommendations.

**Task 7: Interview Questions**

How many digital subscriptions do you currently use?

1. Do you keep track of how much you spend on subscriptions every month?
2. Which devices do you primarily use to access these services?
3. Do you know how much time you spend on each application?
4. Have you ever paid for a subscription that you rarely use?
5. Would you find it useful to see the cost per hour of using a subscription?
6. Would you like recommendations on how to better allocate your digital time?

7. How important is privacy when monitoring device usage?
8. Would you prefer visual dashboards for analyzing your usage patterns?
9. Would notifications about underused subscriptions be helpful?

## Task 8: Identify Requirements

Classify the requirements into

### 1. Functional Requirements

- a. Users must be able to **register and log in to the system**.
- b. Users must be able to **add and manage subscription services**.
- c. The system must **track application usage on Android devices**.
- d. The system must **track application usage on Windows systems**.
- e. The system must **store and process usage data**.
- f. The system must **generate analytics dashboards**.
- g. The system must **calculate cost-per-hour subscription metrics**.
- h. The system must **generate periodic usage reports**.
- i. Users must be able to **set goals and schedules**.

### 2. Non-Functional Requirements (System quality attributes)

- **Security** – User data must be securely stored and protected.
- **Privacy** – Device monitoring must require explicit user consent.
- **Performance** – The system should process analytics efficiently.
- **Scalability** – The system should support multiple users.
- **Usability** – The interface must be simple and intuitive.
- **Reliability** – The system should maintain consistent uptime.

## Task 9: Prepare Use Case List

| Use Case ID | Use Case Name                  | Actor  |
|-------------|--------------------------------|--------|
| UC1         | User Registration              | User   |
| UC2         | User Login                     | User   |
| UC3         | Add Subscription               | User   |
| UC4         | Edit or Delete Subscription    | User   |
| UC5         | Track Device Usage             | System |
| UC6         | View Usage Analytics Dashboard | User   |

|      |                                  |        |
|------|----------------------------------|--------|
| UC7  | Generate Usage Reports           | System |
| UC8  | Receive Optimization Suggestions | User   |
| UC9  | Set Personal Goals               | User   |
| UC10 | Monitor Cross-Device Usage       | System |